FoodHub: Performance Insights and Business Recommendations

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Executive Summary- Key Performance Conclusions

Key Performance Conclusions

- Driven by four main cuisines: American, Japanese, Italian, and Chinese
  - Top performing restaurant: Shake Shack
- Demand Surge: Weekends for **71%** of all orders
- Delivery Gap: Efficiency issues during the week
  - Weekdays: Mean delivery time is **28 minutes** (6 minutes slower)
  - Weekends: Mean delivery time is **22 minutes**
- No Customer Ratings: **39%** of orders have no ratings.
  - Difficult to access the service quality of the resturants

Executive Summary: Actionable Business Insights

- High Value Revenue: Orders over $20 are **29.24%** of the business
  - Orders over $20 have a 25% commission rate compared to 15% for lower orders
- Top Performers: Only 4 restaurants qualify for premium promotional status
  - Premium promotional status: Having >50 ratings and an average rating >4
  - The Meatball Shop, Blue Ribbon Fried Chicken, Shake Shack, and Blue Ribbon Sushi
- Service Time: **10.54%** of orders take over 60 minutes for total fulfillment (preparation plus delivery)
  - This can lead to customer dissatisfaction
- Tiered Commission Structure
  - Zero Revenue **(≤$5):** 0% commission earned on orders $5 or less
  - Standard Value (>$5 to $20): 15% commission.
  - High Value (>$20): 25% commission (29.24% of orders)

Actionable Business Insights

- Incentivize Customer Feedback: Implement a reward program (small discount or loyalty points)
  - This can help reduce the **39%** of unrated orders and improve the quality data
- Optimize weekday deliveries: Investigate why weekday deliveries are **27%** slower than weekends.
  - Look into increasing delivery incentives/staffing during weekday lunch and dinner rushes to improve speed.
- Promote Bundles: Encourage restaurants to offer "Family Packs" or "Executive Combos" priced **above $20**
  - This is will help maximize the number of orders in the 25% commission tier
- New Revenue: Implement a **minimum order value of $5.01** or a "Small Order Fee" for anything $5 or less.
  - This ensures that every delivery made contributes to the company's net revenue.
- Diversify Promotions:
  - **Run targeted ads** for "long tail" cuisines like Middle Eastern or Mediterranean to attract new customer segments

Business Problem Overview

- Context:
  - The restaurant industry in NYC is expanding rapidly, with students and busy professionals increasingly relying on online food delivery services due to hectic lifestyles.
- Challenge:
  - As a food aggregator, FoodHub needs to maintain a competitive edge.
  - The company has collected extensive data on **1,898 orders** but needs to translate this data into a clear understanding of demand patterns and service efficiency.
- Objective:
  - Analyze customer preferences and operational performance to enhance customer experience/ optimize business growth.

Solution Approach

- Data Inspection & Cleaning:
  - Conducted an initial review of the 1,898 orders to understand the data structure (9 variables), check for missing values, and verify data types.
- Univariate Analysis:
  - Analyzed individual variables (cuisine types, order costs, delivery times, and ratings) to identify the baseline distribution of the business.
    - Dominance of American cuisine and the high volume of weekend orders
- Multivariate Analysis:
  - Explored relationships between key metrics, such as how delivery time varies between weekdays and weekends or how cost correlates with cuisine type.

Solution Approach

- Critical Business Metrics
  - Mean delivery times to identify logistics bottlenecks, and total fulfillment time (preparation + delivery) to find the percentage of orders exceeding 60 minutes.
- Targeted Top Performance:
  - Identified "Top Performers" for promotional offers by filtering restaurants that met specific thresholds for both rating volume and average scores.
- Revenue Modeling:
  - Applied tiered commission logic (15% or 25% based on order cost) to determine total net revenue and identify the most profitable order segments.

Data Overview

Data Scope: The dataset tracks the entire order lifecycle, from the restaurant's confirmation to the final drop-off at the customer's door.

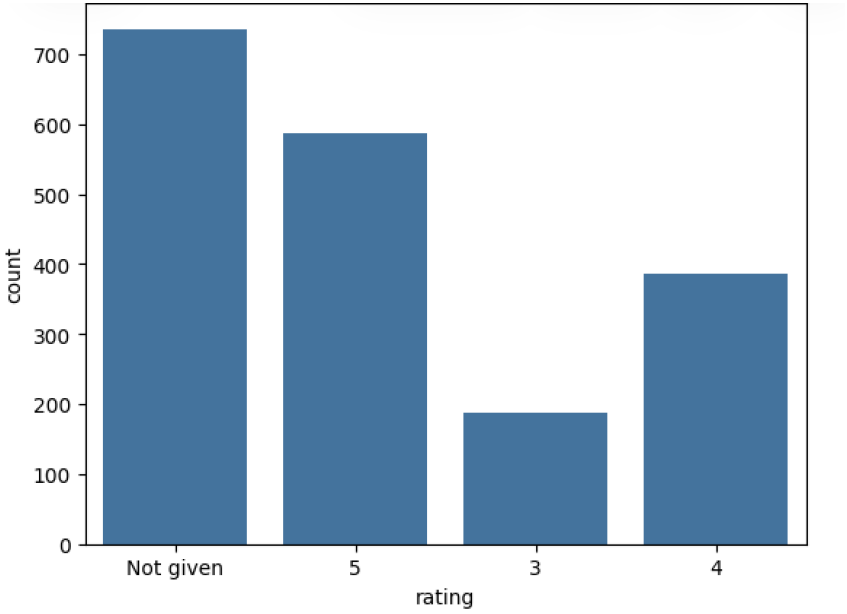
- Total Orders Analyzed: 1,898 unique orders.
- Customer Base: 1,200 unique customers.
- Restaurant Network: 178 unique restaurants.
- Cuisine Diversity: 14 different cuisine types (led by American, Japanese, Italian, and Chinese).

Data Structure

- Dataset Shape: The data consists of 1,898 rows and 9 columns.
- Data Types: The dataset uses a mix of numerical and categorical information:
  - Integers: Order ID, Customer ID, Food Preparation Time, and Delivery Time.
  - Floating point: Cost of the Order.
  - Strings/Objects: Restaurant Name, Cuisine Type, Day of the Week, and Rating.
- No missing values across any of the 9 columns, so there is a complete dataset for analysis.

Operational & Customer Feedback

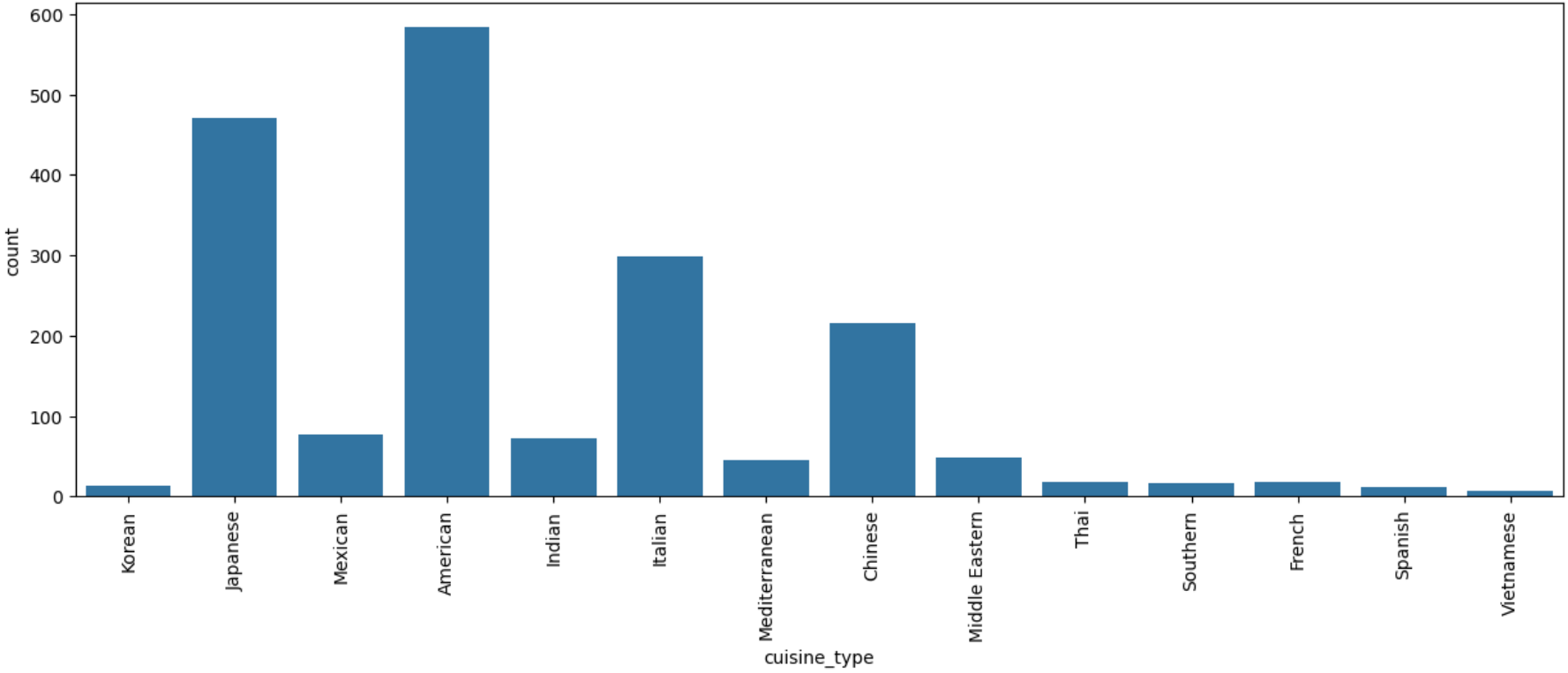
- Food Preparation Benchmarks: Once an order is placed, the time taken by restaurants to prepare the food follows:
  - Minimum Time: 20.0 minutes.
  - Average (Mean) Time: 27.37 minutes.
  - Maximum Time: 35.0 minutes.
- Customer Rating Distribution:
  - Total Unrated Orders:
    - 736 orders are marked as "Not given".
  - For those who did rate:
    - the most common score is a 5 (588 orders)



Univariate Analysis - Categorical Distributions

- Order & Customer Volume:
  - **1,898** unique orders placed by **1,200** unique customers
- Restaurant Network:
  - Orders are spread across **178** unique restaurants
- Cuisine Popularity:
  - There are **14** unique cuisine types. American is the most popular, followed by Japanese, Italian, and Chinese
- Order Timing:
  - Demand is skewed toward the end of the week, with **1,351** orders on weekends compared to **547** on weekdays.
- Customer Feedback:
  - **736 orders (39%)** were not rated. Among those that were rated, a score of 5 is the most frequent.

Univariate Analysis - Categorical Distributions

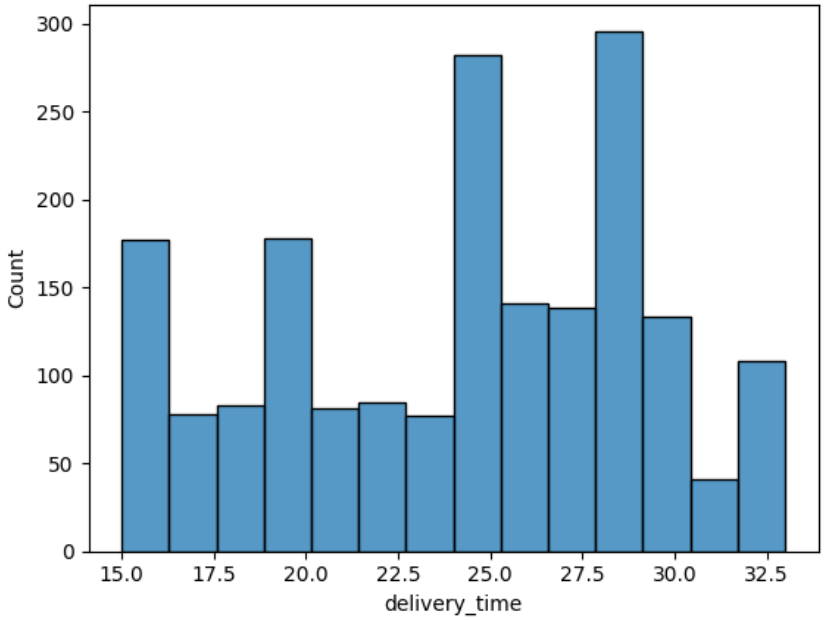


Univariate Analysis - Numerical Metrics

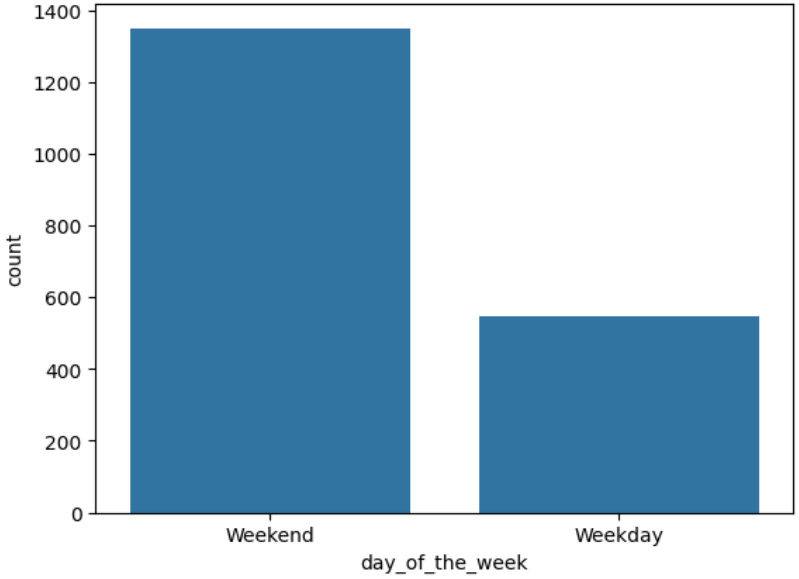
- Order Cost: Costs range from $4.47 to $35.41, with a median cost of $14.14.
  - The distribution shows several peaks, particularly around the $12 and $25 marks.
- Food Preparation Time:
  - This is fairly uniform, ranging from 20 to 35 minutes, with an average of 27.37 minutes
- Delivery Time:
  - Delivery times range from 15 to 33 minutes, with a mean of 24.16 minutes

Univariate Analysis

Histogram for Cost of the Order



Box Plot of Weekend vs. Weekdays



Univariate – Top Performance & Demand

Rank	Restaurant Name	Number of Orders
1	Shake Shack	219
2	The Meatball Shop	132
3	Blue Ribbon Sushi	119
4	Blue Ribbon Fried Chicken	96
5	Parm	68

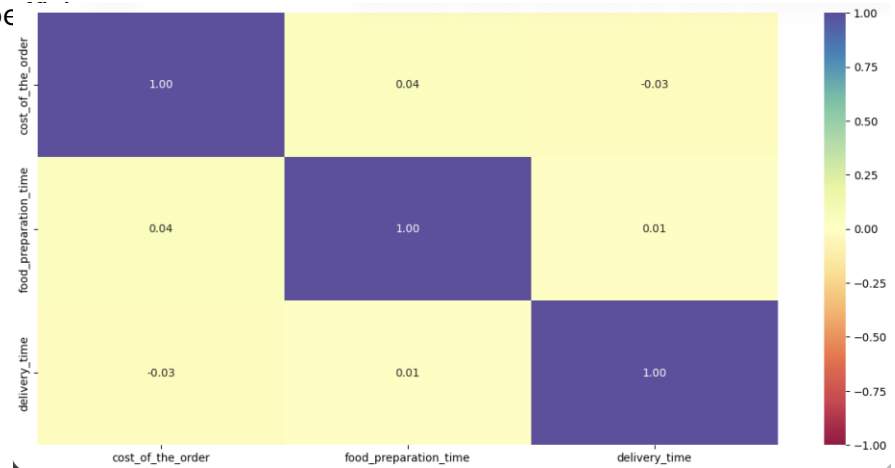
- Weekend Favorites: The most popular cuisine on weekends is American, with 415 orders placed during those two days.

Univariate - Operational & Financial Insights

- High-Value Orders: 555 orders cost more than $20, representing **29.24%** of total orders.
    - These are critical for revenue as they fall into the higher 25% commission bracket.\
- Delivery Efficiency: Across the entire dataset of 1,898 orders, the mean delivery time is 24.16 minutes
- Top 5 Frequent Customers: To drive retention through 20% discount vouchers, the top 5 customers by order count are:
    - ID 52832: 13 orders
    - ID 47440: 10 orders
    - ID 83287: 9 orders
    - ID 250494: 8 orders
    - ID 82041: 7 orders

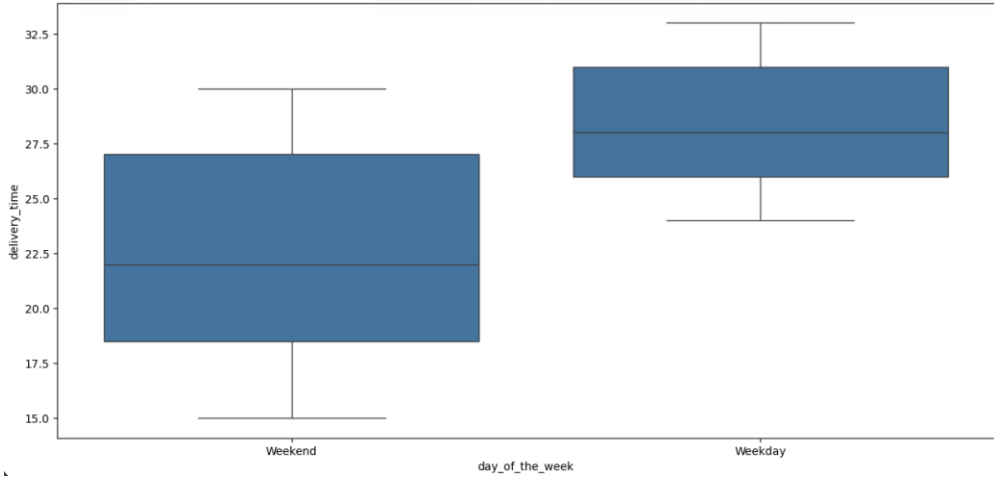
Multivariate Analysis

- Cuisine vs. Cost:
  - Southern and French cuisines tend to have the highest median order costs, while Korean and Vietnamese cuisines are among the least expensive
- Cuisine vs. Preparation Time:
  - Most cuisines show a consistent preparation time of approximately 27 minutes, though Korean food has a slightly lower median preparation time compared to others
- Correlation Heatmap: Shows no significant correlation between order cost, food preparation time, and delivery time, with all correlation coe



Multivariate - Delivery Weekdays vs. Weekends

- Weekday Mean Delivery Time: Approximately 28 minutes
- Weekend Mean Delivery Time: Approximately 22 minutes
- Observation:
  - Weekday deliveries are 27% slower than weekend deliveries, likely due to traffic patterns or courier availability during the work week



Multivariate - Operational Metrics on Ratings

- Rating vs. Delivery Time:
  - Lower ratings (specifically a score of 3) are associated with higher average delivery times, suggesting that late deliveries negatively impact satisfaction
- Rating vs. Preparation Time:
  - Preparation time remains relatively stable across all rating categories (around 27.4 minutes), indicating that the kitchen's speed is less of a factor in low ratings than delivery logistics
- Rating vs. Cost:
  - Orders receiving a 5-star rating tend to have a higher average cost, suggesting that customers are willing to pay more for high-quality experiences

Multivariate - Identifying Strategic Partners

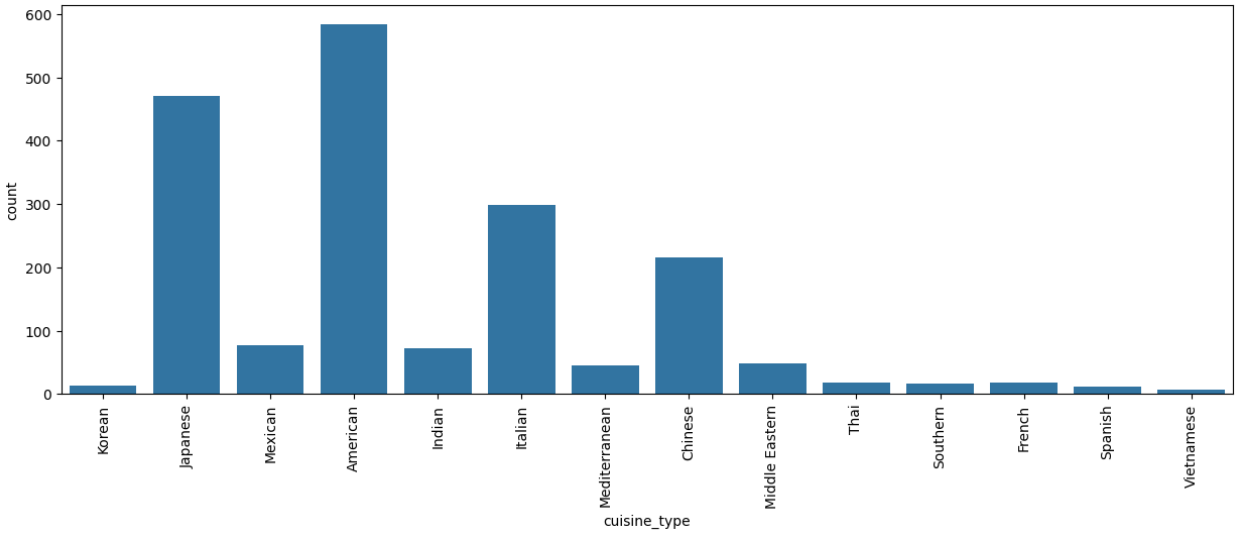
- Top-tier restaurants for a promotional offer based on high volume (>50 ratings) and high quality (Average Rating >4). Only four restaurants currently meet these rigorous standards:
  - The Meatball Shop (Average Rating: 4.51)
  - Blue Ribbon Fried Chicken (Average Rating: 4.33)
  - Shake Shack (Average Rating: 4.28)
  - Blue Ribbon Sushi (Average Rating: 4.22)

Multivariate - Revenue and Service Level Analysis

- Total Net Revenue: By applying a tiered commission FoodHub generated a total net revenue of approximately $6,166.30.
- Fulfillment Efficiency : The "Total Time" (Preparation + Delivery) is a primary indicator of service quality
- Service Risk: 10.54% of all orders take more than 60 minutes to be delivered,
    - This is a significant segment of the business where customer experience is at risk.

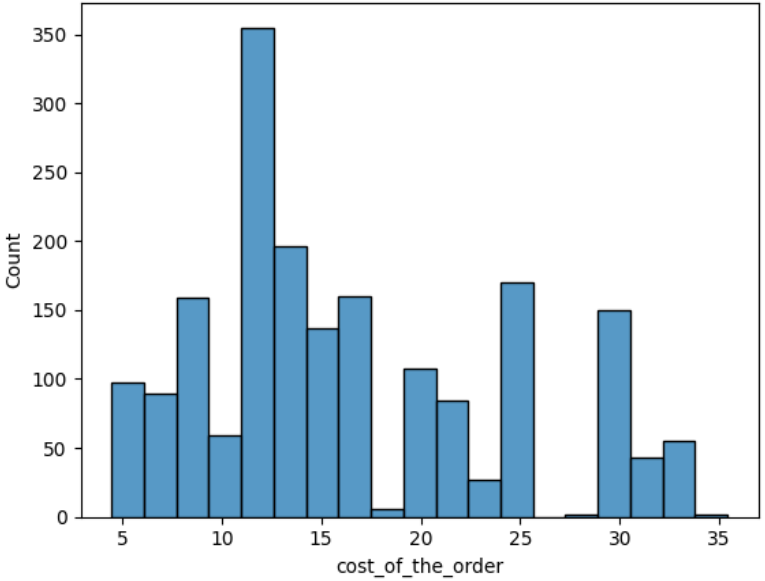
APPENDIX

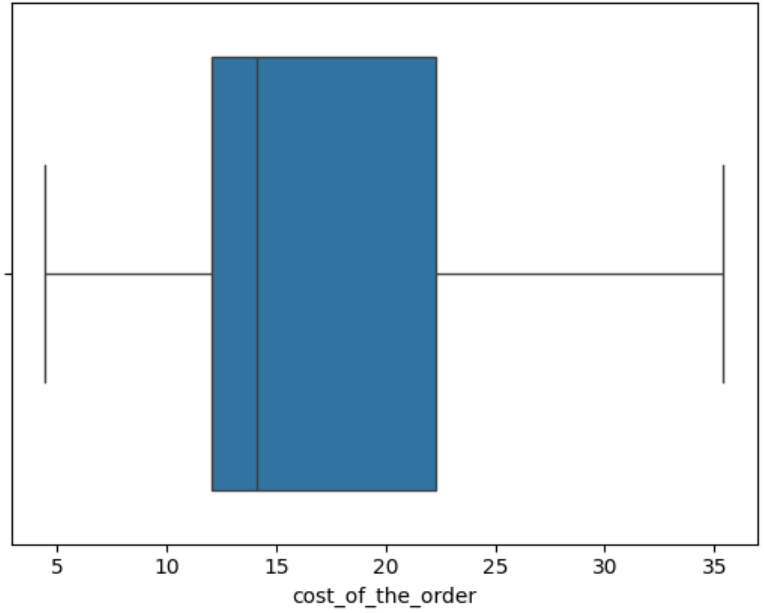
Cuisine Popularity



- American, Japanese, Italian, and Chinese are the core revenue drivers; focus restaurant acquisition and promotions on these Big 4.

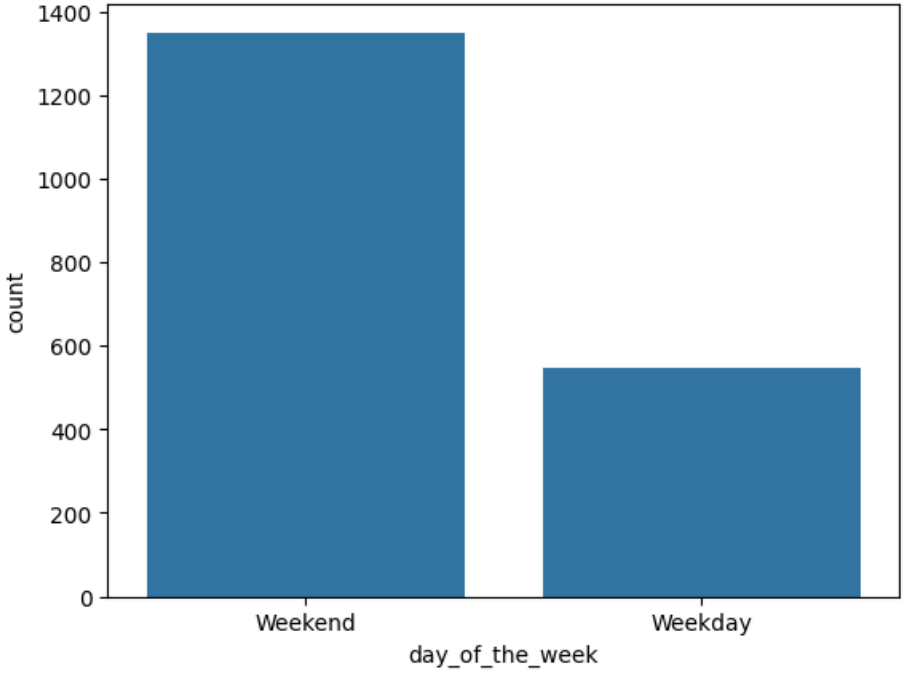
Order Cost:





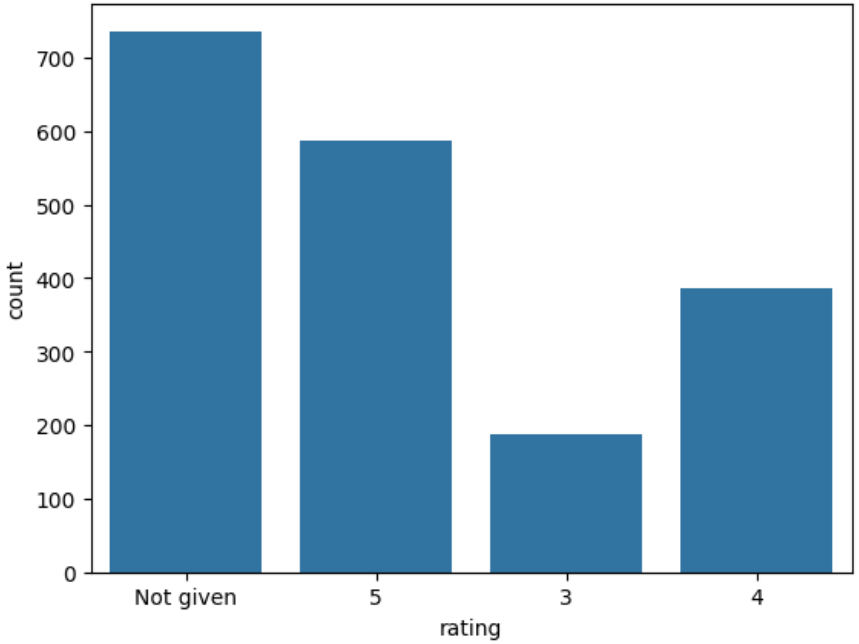
- The median order cost is $14.14, with most transactions falling between $12 and $22. There are no outliers, indicating a stable price environment.

Demand Volume



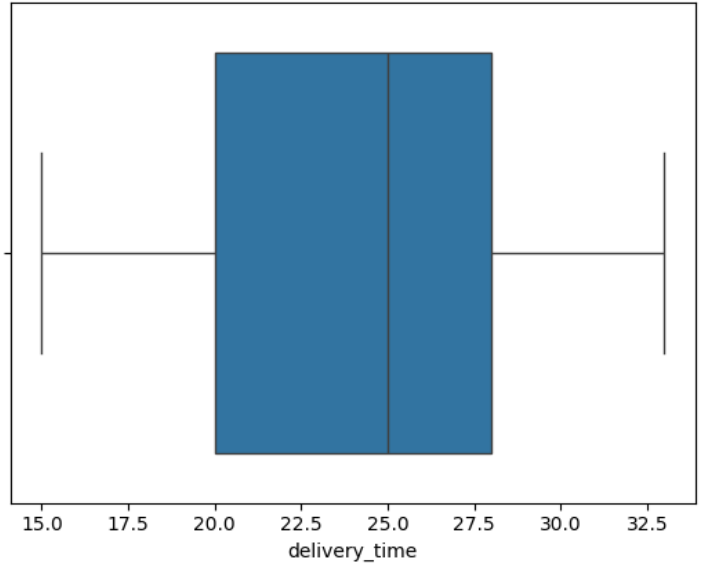
- Demand is heavily skewed toward the weekend (71% of orders). Logistics and staffing must be optimized for this 2.5x volume surge

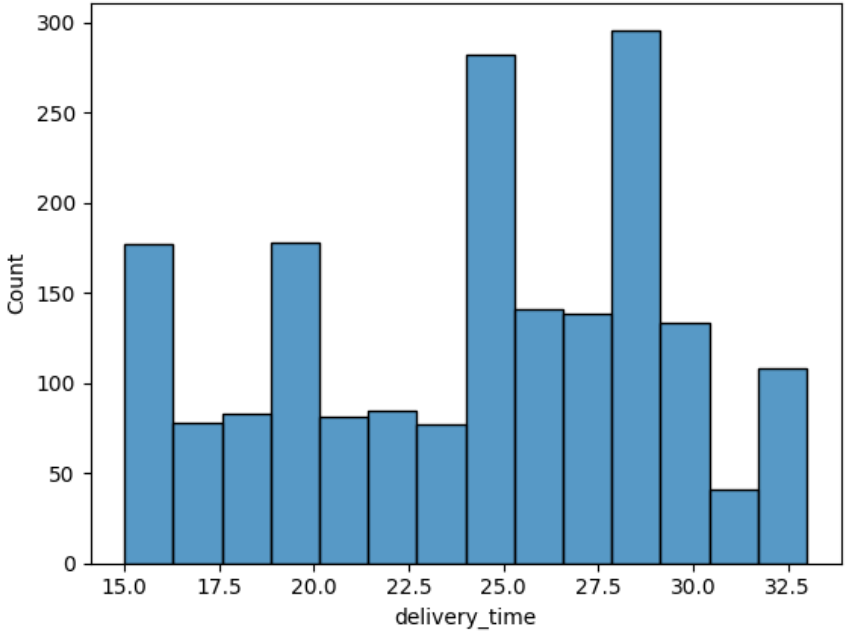
Feedback Engagement



- A critical 39% data gap exists where orders are "Not given" a rating. This limits quality control and requires immediate customer incentives to resolve.

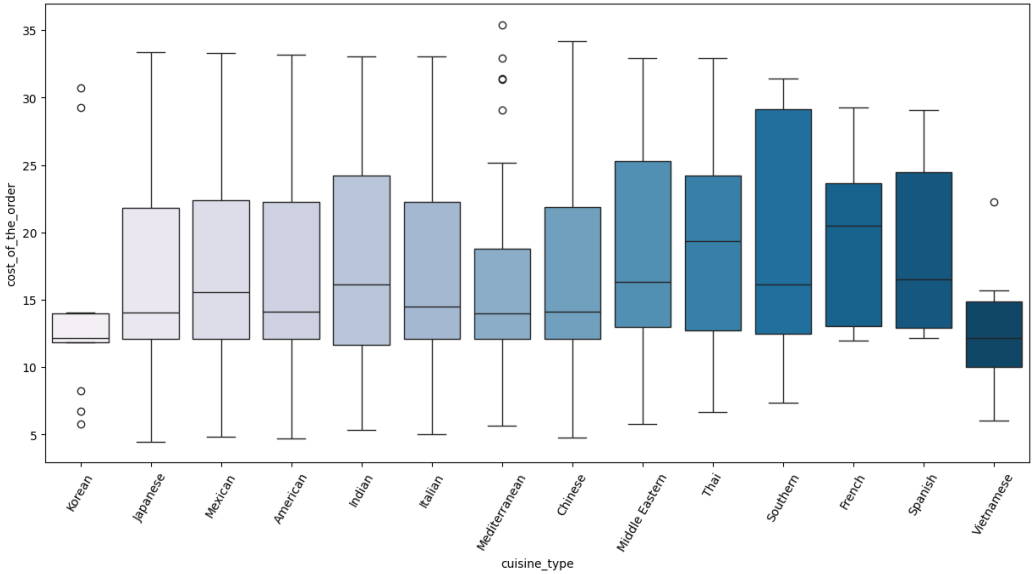
Delivery Baseline





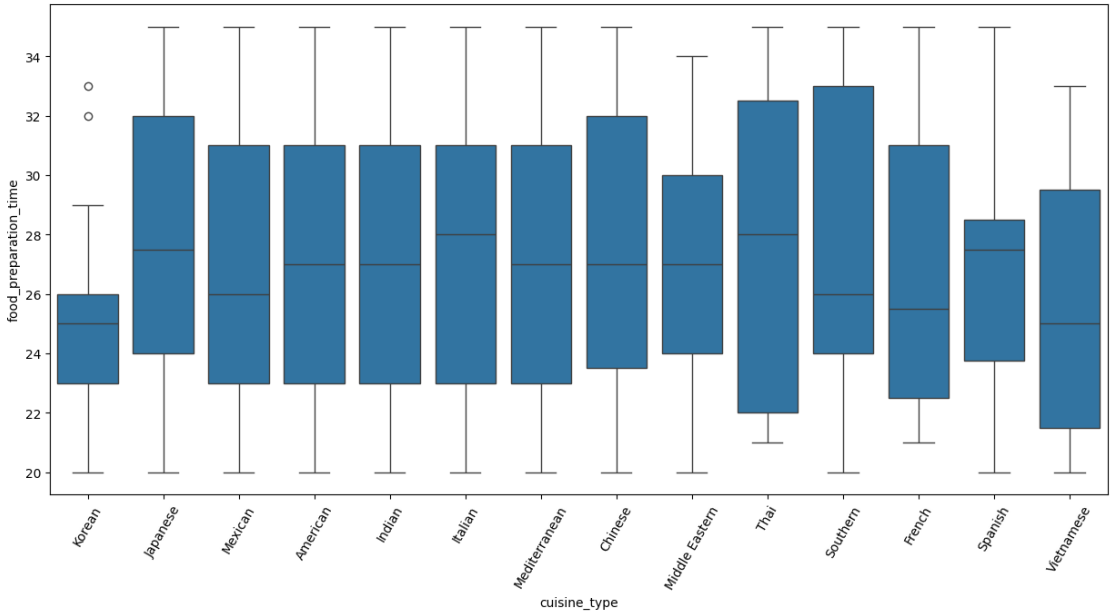
- Average delivery takes 24.16 minutes, ranging from 15 to 33 minutes. This establishes the standard for our baseline service level agreements (SLAs).

Profitability by Cuisine



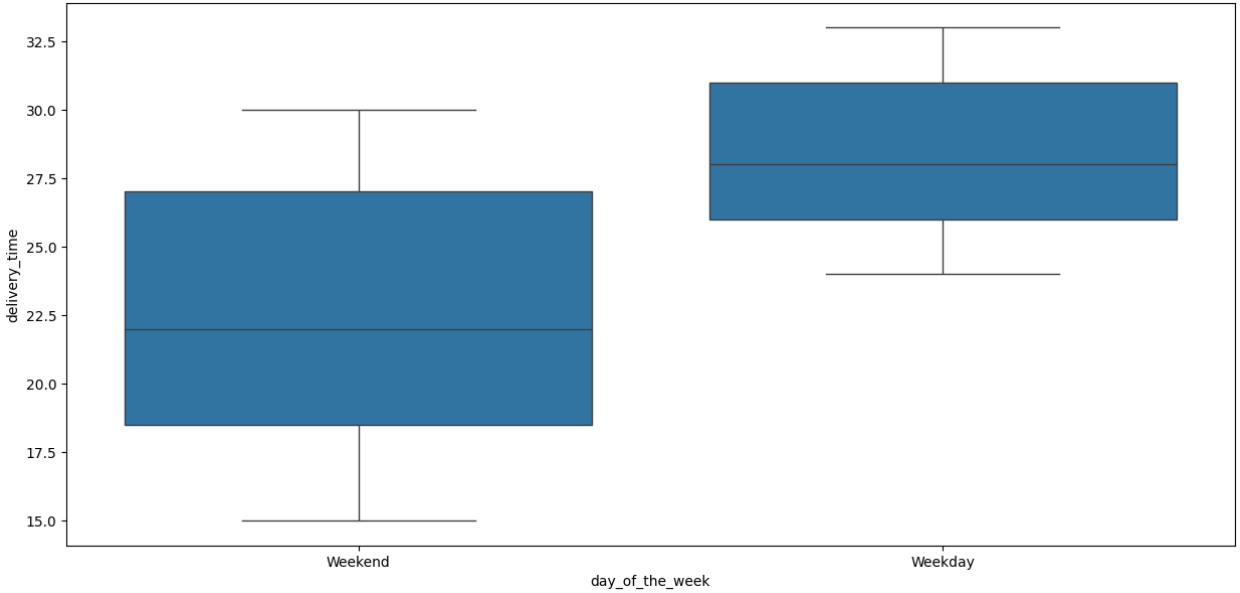
- Southern and French cuisines command the highest median prices, making them primary targets for our higher 25% commission bracket

Kitchen Efficiency



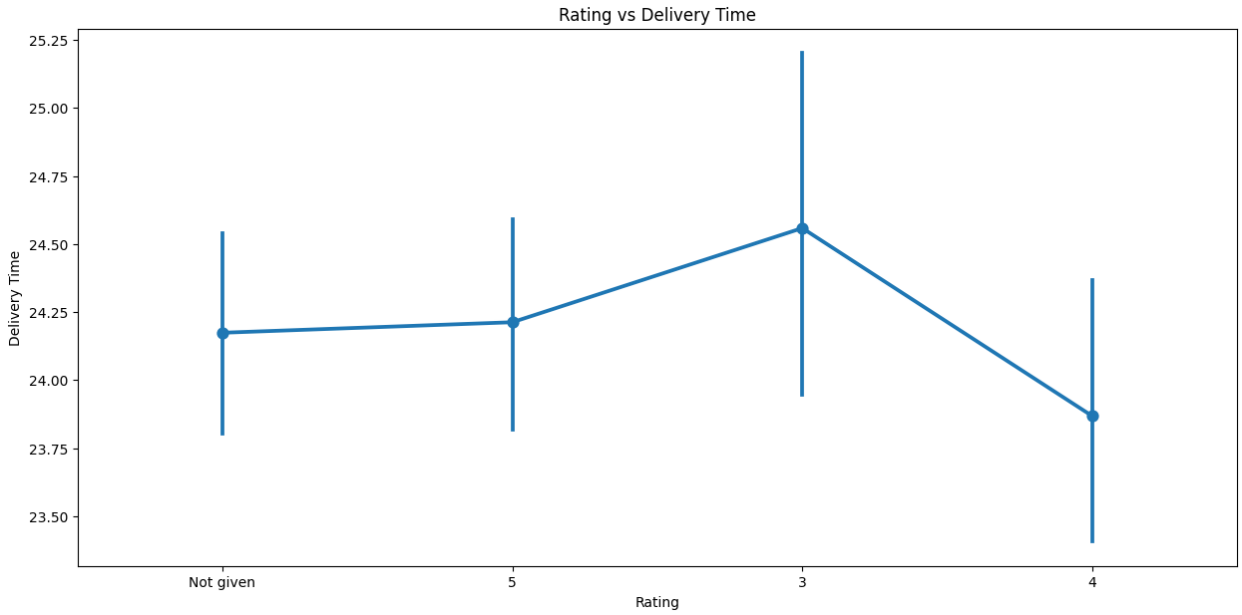
- Preparation time is highly consistent across all cuisines, averaging ~27 minutes. The kitchen is rarely the cause of service delays

Weekday Deliveries



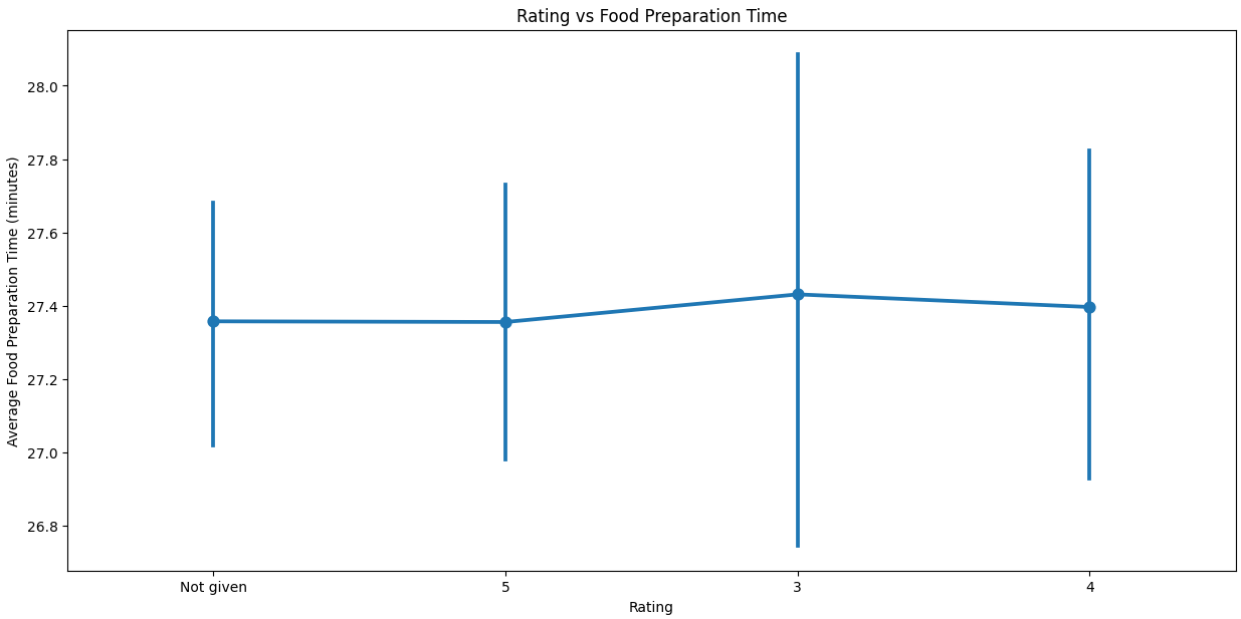
- Weekday deliveries are 27% slower than weekends (28 vs. 22 minutes). This is our most significant operational efficiency gap

Impact of Speed on Satisfaction



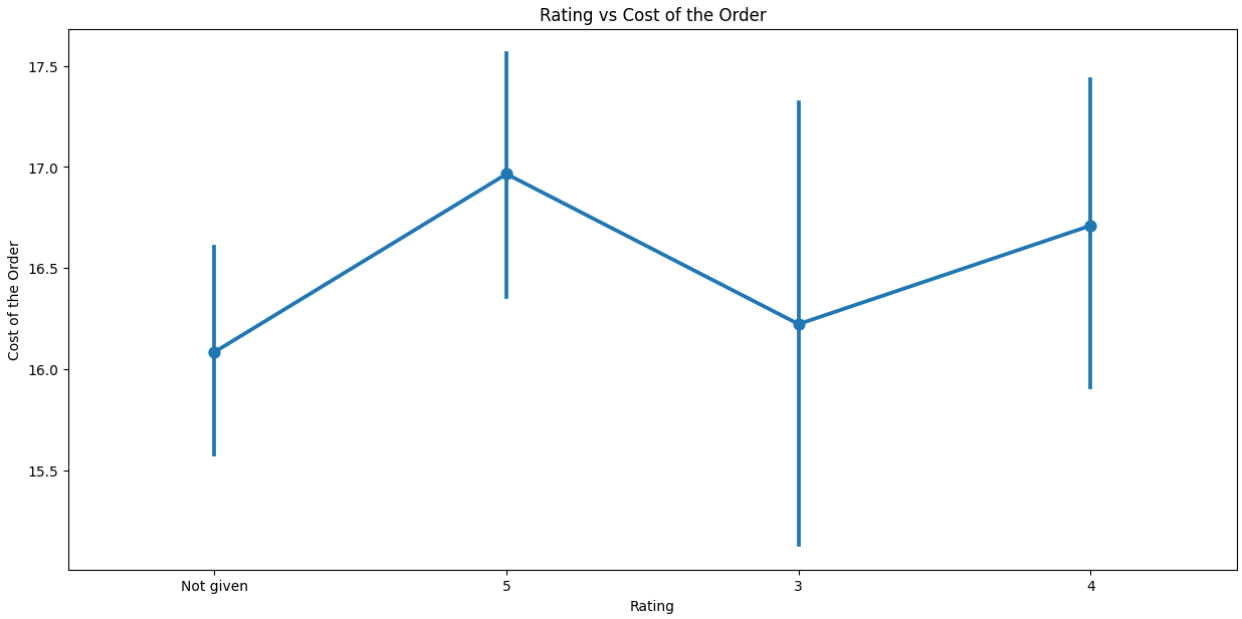
- There is a clear link between delays and dissatisfaction; lower ratings (3-star) correlate with slower delivery times

Quality vs. Preparation



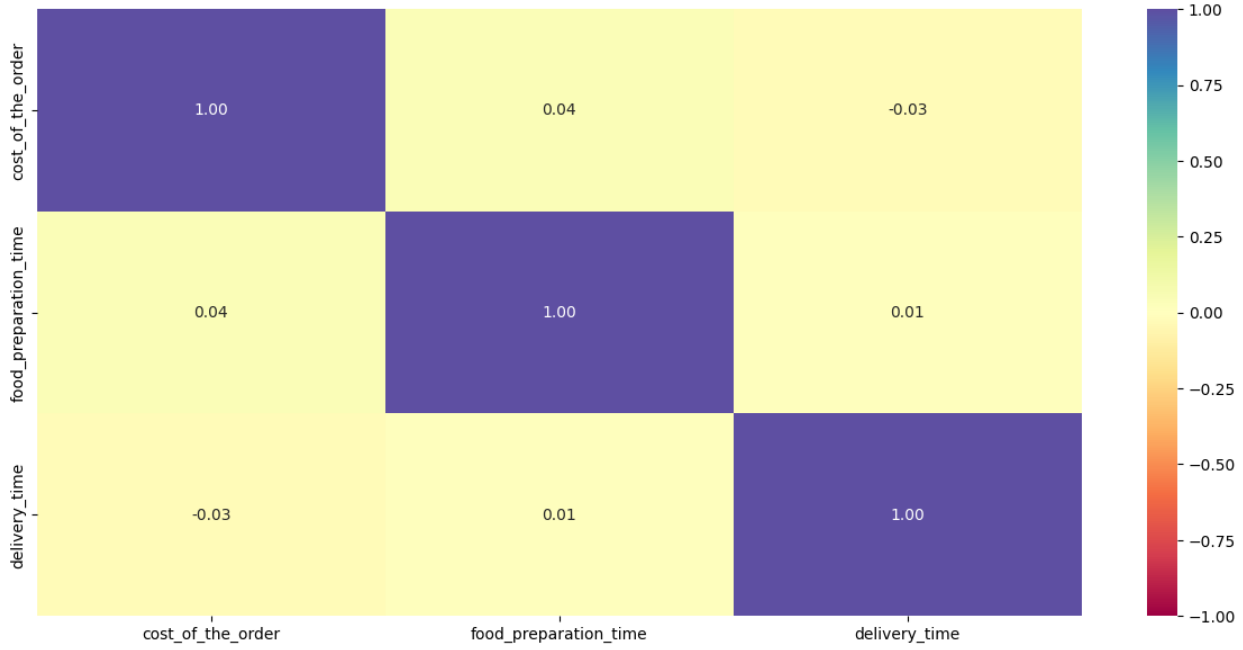
- Preparation speed is identical across all rating levels, proving that delivery logistics—not cooking time—drives customer ratings

Value vs. Satisfaction



- 5-star ratings are associated with higher-cost orders, suggesting that premium customers are the most satisfied with the current end-to-end service.

Variable Independence (Heatmap)



- There is no mathematical correlation between cost, prep time, and delivery time. Our service speed remains consistent regardless of the order's price or complexity.